

Notes: Schwert (JF, 2017)

Municipal Bond Liquidity and Default Risk

Contents

1 Big picture

2 Data and measurement (key objects)

2.1	Transaction data and issuer universe	
2.2	Sample construction	
2.3	Bond characteristics	
2.4	Fundamental prices and yields	
2.5	Key spread concepts	
2.6	Liquidity measures	
2.7	Issuer fiscal condition and external credit measures	

3 Model and empirical specifications (all equations + interpretation)

3.1	Yield decomposition	
3.2	Tax adjustment	
3.3	Liquidity composites from principal components	
3.4	Liquidity pricing regression	
3.5	Transaction-based decomposition	
3.6	CDS-based decomposition	
3.7	Pre-refunding event-study specification	
3.8	Implied default risk premium	

4 Identification and credibility

4.1	What identifies the transaction-based decomposition	
4.2	Why the transaction-based estimates are plausible	
4.3	Why the CDS cross-check matters	
4.4	Why pre-refunding is compelling	
4.5	Robustness	
4.6	What the paper does not pin down perfectly	

5 Tables and figures

5.1	Table I: Sample construction	
5.2	Table II: Liquidity summary statistics and principal components	
5.3	Table III: Liquidity pricing regressions	
5.4	Table IV: Summary statistics for the transaction-based decomposition	
5.5	Figure 1: Time series of decomposed bond spreads	

5.6 Table VI: Determinants of liquidity and default spreads

5.7 Table VII: Comparison of decomposition methods

5.8 Figure 3: Liquidity spreads and the bond-CDS basis

5.9 Table IX: Pre-refunded bonds

5.10 Table X: Implied default risk premium

6 Short summary

7 Conclusion

7.1 Core conclusion (what the paper establishes)

7.2 Economic intuition

7.3 Takeaway

1 Big picture

Question. Once municipal bond yields are adjusted for their tax exemption, how much of the remaining spread reflects *default risk* and how much reflects *liquidity*?

Why this is hard. Municipal bonds differ from Treasuries and corporates along three dimensions at once:

- they are usually exempt from federal and state income taxes;
- they trade over the counter in a retail-dominated market with high transaction costs;
- actual defaults are historically very rare, so raw default frequencies do not obviously explain observed spreads.

Main idea. The paper triangulates the answer with three complementary decompositions:

- a **transaction-based** approach that estimates the liquidity component from bond trading data;
- a **CDS-based** approach that uses state CDS spreads as a direct measure of default risk;
- a **pre-refunding** quasi-natural experiment that studies what happens when a municipal bond is effectively made risk-free until its call date.

Main headline result. After adjusting for taxes, default risk accounts for *most* municipal bond spreads:

- in the full transaction-based sample, the average tax-adjusted spread is about 131 bps, which decomposes into about 101 bps of default and 30 bps of liquidity;
- on the state-CDS subsample, the transaction-based approach attributes about 69% of the spread to default, while the CDS-based approach attributes about 84%;
- around pre-refunding events, spreads fall by about 86.5 bps, implying that default risk explains at least 41% of the pre-event spread even under conservative assumptions.

Main contribution. The paper overturns the view that illiquidity is the dominant non-tax driver of municipal bond spreads. Illiquidity matters, but it is secondary. The first-order pricing force is default risk.

Main mechanism. The large role for default is not driven by high observed default frequencies. Instead, the market assigns a very high *price of default risk*. The implied municipal default risk premium is far larger than in investment-grade corporate bonds.

Credibility strategy. Each decomposition method has weaknesses on its own, but all three point in the same direction. The transaction-based approach could understate absolute illiquidity; the CDS approach only works for a subset of states; and the pre-refunding design delivers only a lower bound. The fact that the answers line up is the paper's main credibility asset.

Policy implication. If policy makers want to reduce state and local borrowing costs, improving secondary-market liquidity is unlikely to move yields much. A larger effect would come from reducing the source of the municipal default risk premium.

Economic takeaway. Municipal borrowing costs are driven much more by priced credit risk than by the high transaction costs that characterize this market.

2 Data and measurement (key objects)

2.1 Transaction data and issuer universe

The core data source is the Municipal Securities Rulemaking Board (MSRB) transaction database, which covers all reported municipal bond trades from January 1998 to June 2015. The raw sample contains roughly 148.5 million trades across 3.1 million distinct bonds.

The paper supplements the MSRB transactions with:

- Bloomberg bond header information, including coupon type, tax status, rating, insurance, callability, and pre-refunding status;
- annual financial data from the U.S. Census Bureau for state and local governments;
- state GDP per capita from the Bureau of Economic Analysis;
- Bloomberg CDS quotes for states with traded CDS contracts, beginning in 2008.

The analysis focuses on **general obligation (GO) bonds** of states and large local governments. Large cities and counties are selected using Census revenue data, with a threshold of average annual revenue above \$50 million.

Interpretation. GO bonds are the right laboratory because they are claims on the issuer's general taxing power, so default risk is naturally issuer-level rather than tied to a specific revenue project.

2.2 Sample construction

The data are heavily cleaned and standardized before estimation. Starting from the full MSRB sample, the paper:

- removes obvious data errors and bonds with too little information;
- restricts attention to matched GO issuers from states, cities, and counties;
- drops variable-rate bonds;
- drops insured bonds;
- drops bonds that are already pre-refunded;
- drops federally taxable bonds and AMT bonds;
- excludes trades in the final year before maturity;
- excludes the first three months after issuance;
- drops callable bonds in the main sample.

The final main sample contains:

- 33,642 bonds;
- 2,526,279 trades.

Interpretation. The restrictions are not cosmetic. They are designed to make yields comparable by removing embedded options, insurance, heterogeneous tax treatment, and the extreme noise in newly issued bonds.

2.3 Bond characteristics

In the final sample:

- mean years to maturity is 4.61;
- mean years since issuance is 5.00;
- mean coupon rate is 4.41%;
- mean issue amount is about \$19.4 million;
- median transaction size is only about \$40,000.

The ratings distribution is extremely skewed toward high quality:

- 29.8% AAA;
- 53.7% AA;
- 15.6% A;
- very few BBB or lower.

State bonds account for 58.8% of trades, cities for 23.6%, and counties for 17.6%.

Interpretation. Even among economically important issuers, the municipal bond market is populated mostly by highly rated securities traded in small sizes by retail investors.

2.4 Fundamental prices and yields

Municipal transactions are noisy because bonds trade infrequently and bid-ask dispersion is large. To smooth away intraday noise, the paper constructs a daily **fundamental price** as follows:

- if both customer sales and customer purchases are observed, take the midpoint of the highest customer-sale price and the lowest customer-purchase price;
- otherwise, use the mean price of interdealer trades;
- if neither is available, omit the observation.

Monthly yields and spreads are then built using the *last* fundamental price of the month.

Interpretation. This is important. The paper does not want retail markups and one-off bargaining outcomes to be mistaken for changes in issuer credit quality.

2.5 Key spread concepts

The basic pricing objects are:

- **Yield to maturity:** the observed municipal yield;
- **Tax-adjusted yield:** the observed tax-exempt yield converted into a taxable-equivalent yield using statutory federal and state tax rates;
- **Tax-adjusted spread:** the tax-adjusted yield minus the interpolated maturity-matched swap rate;
- **Default spread:** the component of the tax-adjusted spread attributed to credit risk;

- **Liquidity spread:** the component attributed to illiquidity.

Interpretation. The paper's empirical object is not the raw municipal yield, but the spread that remains after stripping out the tax advantage and benchmark risk-free rate.

2.6 Liquidity measures

The paper computes six bond-month liquidity metrics:

- Amihud price impact;
- imputed round-trip cost;
- price dispersion;
- Roll measure;
- Amihud risk;
- round-trip cost risk.

It also computes trading-activity variables:

- turnover;
- trades per month;
- interval between trades;
- issuer zero-trade measure;
- dealer-trade turnover.

Median liquidity conditions are poor:

- median Amihud price impact is 3.34% per \$1 million trade;
- median imputed round-trip cost is 0.31%;
- median bond-month has only 11 trades;
- median interval between trades is 7.7 days.

Interpretation. Municipal bonds are undeniably expensive to trade. The paper's claim is not that liquidity is unimportant in levels, but that it explains only a minority of the yield spread.

2.7 Issuer fiscal condition and external credit measures

To connect spreads to fundamentals, the paper uses:

- bond ratings;
- issuer debt-to-revenue;
- state GDP per capita;
- state CDS premia when available.

For the pre-refunding design, the paper separately builds a callable-bond sample of bonds that eventually become pre-refunded. After requiring observations before and after the event, valid

ratings, and at least six months to the call date, the final pre-refunded sample contains 4,636 monthly observations for 1,302 bonds.

Interpretation. These variables play two roles: they help identify default risk directly, and they provide external checks on whether the decomposition behaves as common sense would suggest.

3 Model and empirical specifications (all equations + interpretation)

3.1 Yield decomposition

The paper starts from the idea that a tax-exempt municipal yield reflects taxes, default, and liquidity:

$$y_{i,t} = (1 - \tau_{s,t})(r_t + \gamma_{j,t} + \psi_{i,t}),$$

where:

- i indexes the bond;
- j indexes the issuer;
- s indexes the issuer's state;
- r_t is the risk-free rate;
- $\gamma_{j,t}$ is issuer default risk;
- $\psi_{i,t}$ is bond-specific illiquidity;
- $\tau_{s,t}$ is the marginal tax rate relevant for investors in that state.

Interpretation. The same issuer can have many bonds with the same credit risk but different liquidity. This is the key separation the paper exploits.

3.2 Tax adjustment

The baseline tax adjustment assumes the relevant tax wedge is the top statutory federal times state tax rate:

$$1 - \tau_{s,t} = (1 - \tau_t^{fed})(1 - \tau_{s,t}^{state}).$$

The taxable-equivalent municipal spread is then

$$y_{i,t}^{TA} - r_t = \frac{y_{i,t}}{1 - \tau_{s,t}} - r_t = \gamma_{j,t} + \psi_{i,t}.$$

Interpretation. Once the tax wedge is removed, the remaining spread should consist only of default and liquidity. The whole empirical project is to split that object into its two parts.

3.3 Liquidity composites from principal components

The first principal component of the six liquidity measures is approximated by

$$\lambda_{i,t} = \sum_{k=1}^6 \frac{L_{i,t}^k - \mu_k}{\sigma_k},$$

where L^k runs over the six liquidity metrics and μ_k, σ_k are full-sample means and standard deviations.

The paper also defines a trading-activity composite,

$$\lambda_{i,t}^T = \sum_{k=1}^3 \frac{T_{i,t}^k - \mu_k}{\sigma_k},$$

which is built from the variables that load most heavily on the second principal component: trade interval, issuer zero-trade, and minus trades per month.

Interpretation. $\lambda_{i,t}$ is meant to capture *transaction-cost liquidity*, while $\lambda_{i,t}^T$ is meant to capture *trading intensity or market activity*.

3.4 Liquidity pricing regression

The paper relates tax-adjusted spreads to liquidity, trading activity, and ratings:

$$y_{i,t}^{TA} - r_t = \alpha + \beta_\lambda \lambda_{i,t} + \beta_{\lambda^T} \lambda_{i,t}^T + \beta_{Turnover} Z(Turnover_{i,t}) + Rating_{i,t} + \varepsilon_{i,t}.$$

The baseline evidence is:

- the liquidity composite λ is strongly priced;
- trading-activity measures add little explanatory power once λ is included;
- with month fixed effects, turnover becomes insignificant.

Interpretation. The market prices expected trading costs more than sheer trading frequency. This already hints that illiquidity is a part of spreads, but not necessarily the main part.

3.5 Transaction-based decomposition

For the actual decomposition, the paper estimates month-specific liquidity loadings:

$$y_{i,t}^{TA} - r_t = \alpha + \beta_t^\lambda \lambda_{i,t} + Rating_{i,t} + \varepsilon_{i,t}.$$

It then benchmarks each bond against a very liquid municipal bond, defined as the 1st percentile of the liquidity distribution:

$$\psi_{i,t} = \beta_t^\lambda (\lambda_{i,t} - \lambda_{1p}).$$

Finally, the default component is the residual:

$$\gamma_{i,t} = y_{i,t}^{TA} - r_t - \psi_{i,t}.$$

Interpretation. This procedure attributes the portion of spreads that covaries with observed illiquidity to liquidity, and whatever remains to default risk.

3.6 CDS-based decomposition

The second approach follows the logic of Longstaff, Mithal, and Neis (2005). For each issuer-month with CDS data, the paper first estimates a five-year municipal bond spread by regressing individual

bond spreads on time to maturity using bonds in a one- to nine-year window, and then taking the fitted value at five years.

The decomposition is then:

$$\text{Default Spread}_{j,t} = CDS_{j,t}, \quad \text{Liquidity Spread}_{j,t} = \text{Spread}_{j,t}^{TA} - CDS_{j,t}.$$

Interpretation. CDS gives an external market-based measure of default risk. If the CDS premium is close to the transaction-based default spread, confidence in the main result rises sharply.

3.7 Pre-refunding event-study specification

The third approach exploits the fact that pre-refunding effectively makes a bond risk-free until the call date. The main regression is

$$y_{i,t}^{TA} - r_t = \alpha + \beta \text{PreRefunded}_{i,t} + \text{Rating}_{i,t} + \text{Controls}_{i,t} + \delta_i + \delta_t + \varepsilon_{i,t},$$

where:

- $\text{PreRefunded}_{i,t}$ equals one after the pre-refunding event;
- controls include years to the call date and log issue size;
- δ_i and δ_t are bond and month fixed effects.

Interpretation. β measures the within-bond drop in spreads when default risk is removed. Because pre-refunding may make a bond *less* liquid, this estimate is interpreted as a *lower bound* on the default component.

3.8 Implied default risk premium

To translate default spreads into a risk-premium multiple, the paper uses

$$\gamma = q \times LGD,$$

where γ is the default spread, q is the risk-neutral annual default probability, and LGD is expected loss given default.

The paper then compares q with the physical default probability p , using q/p as a default risk premium multiple.

Interpretation. If actual municipal defaults are extremely rare but q inferred from spreads is much larger than p , then spreads reflect a large price of risk rather than merely large expected losses.

4 Identification and credibility

4.1 What identifies the transaction-based decomposition

The transaction-based approach identifies liquidity from the cross-sectional relation between tax-adjusted spreads and the liquidity composite λ , conditional on credit ratings. The basic logic is:

- if two bonds have similar credit but one is harder to trade, the harder-to-trade bond should have a higher spread;
- the estimated spread loading on liquidity is then converted into a dollar liquidity spread.

Potential concern. Ratings are imperfect controls for default risk, so residual credit differences could contaminate the liquidity coefficient.

4.2 Why the transaction-based estimates are plausible

The paper checks whether the decomposed pieces line up with variables that should matter for default but not much for liquidity. In the regressions of Table VI:

- issuer debt-to-revenue loads mainly on the default spread;
- state GDP per capita loads mainly on the default spread;
- rating indicators load much more strongly on default than on liquidity;
- duration and bond age affect both components, which is sensible because longer and older bonds can be both riskier and harder to trade.

Interpretation. This is an important “smell test.” If fiscal variables had loaded mostly on the liquidity component, the decomposition would be hard to believe.

4.3 Why the CDS cross-check matters

The CDS-based approach is attractive because it measures default with a different asset class. The main advantages are:

- it does not infer default mechanically from bond illiquidity;
- CDS are generally less exposed to the retail trading frictions that affect municipal bonds.

Limitation. State CDS exist only for a subset of issuers and only from 2008 onward. Moreover, CDS themselves may contain some liquidity premium.

Even so, the paper argues that this concern is not fatal. If some part of state CDS premia reflects CDS illiquidity, then the CDS-based decomposition would *overstate* liquidity and *understate* the role of default. Yet default still accounts for most of the spread.

4.4 Why pre-refunding is compelling

Pre-refunding is the cleanest design in the paper because it studies a within-bond change in default risk:

- the same bond is observed before and after becoming effectively risk-free until the call date;
- bond fixed effects absorb time-invariant bond mispricing;
- month fixed effects absorb aggregate market conditions.

Limitation. Pre-refunding is not random, and post-event liquidity can worsen. Both considerations make the estimated spread decline a lower bound, not a point-identified full effect.

4.5 Robustness

The main proportional decomposition survives a long list of robustness checks:

- alternative tax adjustments based on Green (1993) and Longstaff (2011);
- alternative risk-free benchmark using Treasury curves instead of swaps;
- alternative samples including callable bonds, insured bonds, interdealer trades only, and state bonds only.

Interpretation. The exact level of tax-adjusted spreads changes across specifications, but the paper’s main conclusion — default risk dominates liquidity — is stable.

4.6 What the paper does not pin down perfectly

The paper is careful about what remains uncertain:

- the correct marginal tax rate embedded in municipal yields;
- the absolute liquidity level of even the *most liquid* municipal bonds relative to swaps or Treasuries;
- the fundamental source of the very large municipal default risk premium.

Interpretation. The paper is stronger on the relative decomposition of spreads than on the deeper structural origin of the risk premium.

5 Tables and figures

5.1 Table I: Sample construction

Read.

- The raw MSRB data contain more than 148 million trades and over 3 million bonds.
- After cleaning and standardizing the sample, the paper ends up with 33,642 bonds and about 2.53 million trades.
- The final sample is mostly high-grade bonds with modest issue sizes and very small transaction sizes.
- The median trade is only about \$40,000, underscoring the retail character of this market.

Economic message. The paper is not studying all municipals indiscriminately. It builds a standardized sample in which differences in spreads are less likely to reflect options, insurance, or weird tax treatment.

5.2 Table II: Liquidity summary statistics and principal components

Read.

- The bond market is clearly illiquid: price impact and round-trip cost measures are large, and trades are infrequent.
- The first principal component loads roughly evenly across the liquidity metrics.

Table I
Sample Construction

This table summarizes the construction of the municipal bond transaction sample. Panel A describes the steps involved in cleaning the transaction data. The Census match step restricts the sample to general obligation bonds from states, cities, and counties with annual revenues over \$50 million. The removal of data errors involves dropping bonds with missing information in the MSRB data, coupons greater than 20%, maturities over 100 years, and fewer than 10 trades in the sample period, as well as individual trades occurring at prices below 50 and above 150. See Section I for a description of the remaining steps of the sample construction. Panel B describes the distribution of bond characteristics and transaction sizes. Panel C reports the proportion of transactions falling under different categories.

Panel A: Steps to Cleaning Transaction Data			
Cleaning Step	Bonds	Trades	
Full MSRB sample	3,105,459	148,549,596	
Remove data errors	1,319,944	129,850,645	
Census match	133,079	16,906,318	
Drop variable rate bonds	132,397	16,691,182	
Drop insured bonds	77,442	10,527,863	
Drop pre-refunded bonds	67,818	9,277,663	
Drop taxable and AMT bonds	64,381	8,460,582	
Exclude last year before maturity	63,533	7,868,733	
Exclude three months after issuance	61,283	6,137,212	
Drop callable bonds	33,642	2,526,279	
Final sample	33,642	2,526,279	

Panel B: Bond Characteristics							
	Mean	StDev	P_1	P_{10}	P_{50}	P_{90}	P_{99}
Years to Maturity	4.61	2.64	1.06	1.58	4.15	8.19	12.5
Years since Issuance	5.00	4.46	0.31	0.92	3.82	10.7	21.3
Coupon Rate (%)	4.41	1.60	0	2.75	5.00	5.50	7.50
Issue Amount (\$MM)	19.4	28.5	0.37	1.53	10.0	48.1	137
Transaction Amount (\$000s)	228	1101	5.00	10.0	40.0	300	4715

Panel C: Categorical Bond Characteristics							
	AAA	AA	A	BBB	$\leq$ BB	NR	
Rating Category (%)	29.8	53.7	15.6	0.63	0.06	0.04	
	CA	NY	MD	NC	MA	TX	OH
State (%)	12.2	11.4	6.74	5.81	5.67	5.36	5.18
	State	City	County				
Government Level (%)	58.8	23.6	17.6				

Table II
Liquidity Summary Statistics and Principal Components Analysis

This table reports statistics on the liquidity and trading activity variables, which are described in the Appendix. Statistics are based on 40,495 monthly observations for individual bonds. Panel A reports summary statistics. Panel B reports the results of a principal components analysis. The principal components analysis is performed on the correlation matrix, which accounts for different variances among the variables. The reported values are standardized scoring coefficients scaled up by a factor of 100.

Panel A: Summary Statistics							
	Mean	StDev	P_1	P_{10}	P_{50}	P_{90}	P_{99}
<i>Liquidity Variables</i>							
Amihud Measure (% per \$million)	9.79	19.7	0.02	0.31	3.34	23.2	122
Imputed Round-Trip Cost (%)	0.45	0.46	0	0.05	0.31	0.99	2.42
Price Dispersion	0.31	0.25	0.02	0.07	0.25	0.62	1.21
Roll Measure	1.01	0.79	0	0.21	0.82	2.03	3.92
Amihud Risk (%)	12.8	23.0	0.02	0.33	4.47	32.1	146
Round-Trip Cost Risk (%)	10.5	56.8	0.02	0.31	3.24	22.4	109
<i>Trading Activity Variables</i>							
Turnover (%)	22.0	51.3	0.30	1.03	5.63	53.8	256
Trades per Month	14.6	12.3	5	6	11	25	62
Trade Interval	16.1	31.5	1.63	3.09	7.67	32.2	146
Issuer Zero Trade	0.38	0.15	0.23	0.27	0.33	0.63	0.90
Turnover (Dealer Trades) (%)	8.63	25.4	0.05	0.23	1.67	18.9	125

Panel B: Principal Components Analysis					
	1PC	2PC	3PC	4PC	5PC
Amihud Measure	24.1	13.6	0.26	25.2	-1.84
Imputed Round-Trip Cost	23.4	-2.93	11.0	-29.4	-3.76
Price Dispersion	24.2	-6.27	9.64	-32.0	-1.82
Roll Measure	20.3	-6.63	6.46	-35.3	-3.84
Amihud Risk	22.3	4.80	2.65	25.4	-2.33
Round-Trip Cost Risk	14.0	14.0	2.74	71.4	7.91
Turnover	-7.21	3.68	70.9	13.6	-11.1
Trades per Month	-1.51	-44.4	48.8	7.66	13.5
Trade Interval	-2.21	46.9	20.5	-21.0	85.9
Issuer Zero Trade	-4.45	48.0	21.4	-14.8	-73.2
Cumulative % Explained	34.5%	48.9%	60.6%	70.2%	77.8%

- The second principal component captures low trading frequency, while the third is mostly turnover.
- This supports the use of a single liquidity composite λ for the main decomposition.

Economic message. The decomposition is not hanging on one ad hoc liquidity variable. It is built from a broad set of trading-cost proxies that move together.

5.3 Table III: Liquidity pricing regressions

Read.

- The normalized liquidity composite $Z(\lambda)$ enters positively and very significantly, with a coefficient around 34 bps in the main specifications.
- Adding $Z(\lambda)$ raises R^2 from about 0.17 to around 0.28 to 0.29.
- The trading-activity composite is smaller and negative, and turnover becomes insignificant once month fixed effects are included.
- With month fixed effects, the loading on $Z(\lambda)$ remains strong at about 29 bps.

Economic message. Liquidity is priced. But the table also suggests that liquidity cannot plausibly explain the *entire* municipal spread, which is why the residual default component remains economically large.

5.4 Table IV: Summary statistics for the transaction-based decomposition

Read.

- The average tax-adjusted spread is 131 bps in the full sample.

Table III
Liquidity Pricing Regressions

This table reports regressions of tax-adjusted yield spreads on liquidity and trading activity factors. The dependent variable is the tax-adjusted spread, which equals the tax-adjusted yield to maturity minus the interpolated maturity-matched swap rate. The liquidity variable $Z(\lambda)$ is the normalized sum of the six normalized liquidity measures and is designed to mimic the first principal component of the liquidity and trading activity variables. The trading activity variable $Z(\lambda^T)$ is the normalized sum of normalized issuer zero-trade days and normalized trading interval minus normalized trades per month, approximating the second principal component. $Z(Turnover)$ is the normalized turnover and proxies for the third principal component. The rating variables are dummies equal to one if the bond is in that rating category. The missing category includes the nonrated bonds. Month fixed effects are included in the last specification. t -statistics based on standard errors clustered by bond and month are reported in brackets. *, **, and *** indicate that the corresponding p -values are less than 0.05, 0.01, and 0.001, respectively.

Tax-Adjusted Spread	(1)	(2)	(3)	(4)	(5)
$Z(\lambda_{i,t})$		34.3*** [28.6]	34.4*** [29.2]	34.9*** [28.5]	29.4*** [35.1]
$Z(\lambda^T_{i,t})$			-7.40*** [-4.97]	-7.23*** [-4.89]	-4.18*** [-4.73]
$Z(Turnover)$				2.45** [3.32]	-0.13 [-0.33]
AAA-Rated	-245*** [-5.73]	-175* [-2.41]	-175* [-2.42]	-174* [-2.41]	-173** [-3.01]
AA-Rated	-222*** [-5.21]	-155* [-2.15]	-157* [-2.18]	-156* [-2.16]	-152** [-2.65]
A-Rated	-152*** [-3.59]	-95.4 [-1.33]	-102 [-1.42]	-101 [-1.40]	-97.3 [-1.71]
BBB-Rated	-141** [-3.15]	-93.7 [-1.28]	-90.5 [-1.23]	-96.1 [-1.23]	-96.1 [-1.64]
BB-to-D-Rated	74.0 [1.73]	103 [1.42]	102 [1.41]	103 [1.42]	109 [1.90]
Constant	345*** [8.06]	284*** [3.92]	286*** [3.95]	285*** [3.94]	318*** [5.55]
Month FEs					X
R^2	0.17	0.28	0.29	0.29	0.51
Observations	37,931	37,931	37,931	37,931	37,931

Table IV

Summary Statistics for Transaction-Based Decomposition

This table reports summary statistics for the transaction-based bond spread decomposition. Observations are reported at the bond-month level. *Yield* is yield to maturity. *Tax-Adjusted Yield* is the yield adjusted upward with the top statutory income tax rate as in equation (3). *Tax-Adjusted Spread* is the tax-adjusted yield minus the interpolated maturity-matched swap rate. *Default Spread* and *Liquidity Spread* are the respective components of the tax-adjusted spread resulting from the transaction-based procedure outlined in Section I.I.C. *Default/Tax-Adj.* is the ratio of the default spread to the tax-adjusted spread.

	Mean	StDev	p_1	p_{10}	p_{50}	p_{90}	p_{99}
<i>Full Sample (38,091 Observations)</i>							
Yield (%)	2.57	1.31	0.42	0.84	2.56	4.27	5.34
Tax-Adjusted Yield (%)	4.32	2.24	0.73	1.45	4.24	7.34	9.41
Tax-Adjusted Spread (bps)	131	83.1	-3.77	40.8	116	241	419
Default Spread (bps)	101	75.8	-29.3	19.2	88.6	198	355
Liquidity Spread (bps)	30.0	27.3	0	5.13	22.4	64.3	129
Default/Tax-Adj.	0.74	0.34	-0.94	0.43	0.81	0.95	2.23
<i>1998 to 2006 (13,047 Observations)</i>							
Yield (%)	3.65	0.91	1.40	2.38	3.68	4.86	5.38
Tax-Adjusted Yield (%)	6.21	1.69	2.29	3.94	6.15	8.56	9.48
Tax-Adjusted Spread (bps)	130	70.1	9.19	46.9	121	223	336
Default Spread (bps)	101	62.9	-17.4	26.4	95.4	184	273
Liquidity Spread (bps)	28.2	28.1	0	4.14	19.6	63.0	136
Default/Tax-Adj.	0.76	0.29	-0.49	0.48	0.83	0.96	1.00
<i>2007 to 2009 (7,141 Observations)</i>							
Yield (%)	3.14	0.87	0.98	1.84	3.30	4.07	4.98
Tax-Adjusted Yield (%)	5.17	1.45	1.60	3.03	5.43	6.72	8.26
Tax-Adjusted Spread (bps)	164	102	-4.97	58.6	138	324	419
Default Spread (bps)	133	93.8	-31.6	32.0	112	277	377
Liquidity Spread (bps)	31.1	28.2	0	4.76	23.6	66.9	135
Default/Tax-Adj.	0.78	0.33	-0.92	0.53	0.85	0.96	2.23
<i>2010 to 2015 (17,903 Observations)</i>							
Yield (%)	1.55	0.85	0.41	0.64	1.35	2.73	4.15
Tax-Adjusted Yield (%)	2.61	1.39	0.71	1.11	2.30	4.54	6.76
Tax-Adjusted Spread (bps)	118	79.8	-4.97	33.0	103	222	413
Default Spread (bps)	87.5	72.4	-34.9	12.3	75.1	178	345
Liquidity Spread (bps)	30.8	26.1	0.48	6.12	24.0	63.9	122
Default/Tax-Adj.	0.70	0.37	-0.94	0.36	0.78	0.93	2.23

- The transaction-based decomposition attributes 101 bps to default and 30 bps to liquidity.
- The average default-share ratio is 0.74, while the median is 0.81.
- In the crisis subperiod 2007 to 2009, default spreads rise to 133 bps, while liquidity spreads move only modestly to 31 bps.

Economic message. This is the paper's core result in one table: default risk, not liquidity, is the main driver of tax-adjusted municipal spreads.

5.5 Figure 1: Time series of decomposed bond spreads

Read.

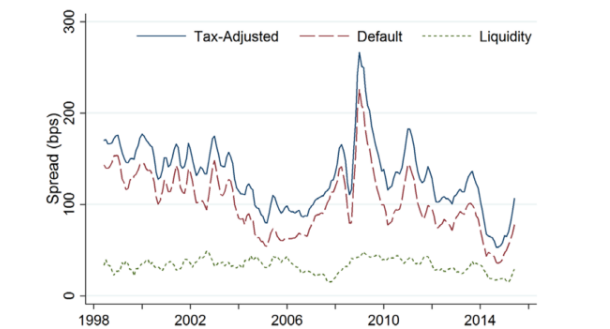
- Tax-adjusted spreads and default spreads move closely together over time.
- The huge run-up in spreads during 2008 is mirrored primarily in the default component.
- Liquidity spreads are much lower in level and much flatter over time.
- After the crisis, both total spreads and default spreads decline steadily.

Economic message. The crisis was not mainly a municipal liquidity event. The time-series behavior points much more strongly to repricing of credit risk.

5.6 Table VI: Determinants of liquidity and default spreads

Read.

- Issuer debt-to-revenue and lower state GDP per capita increase tax-adjusted spreads.
- Those fiscal variables load mainly on the *default* component, not the liquidity component.
- Higher ratings reduce default spreads much more than liquidity spreads.



1. Time series of decomposed bond spreads. This figure presents time-series plots of decomposed bond spreads. Each line presents equal-weighted averages over four-month rolling windows. The solid line is the tax-adjusted spread over the swap rate, the dashed line is the default spread, and the dotted line is the liquidity spread. (Color figure can be viewed at [wileyonlinelibrary.com](#))

Table VI

Determinants of Liquidity and Default Spreads

This table reports panel regressions of bond spreads on bond and issuer characteristics. *Tax-Adjusted Spread* is the tax-adjusted yield minus the interpolated maturity-matched swap rate. *Default Spread* and *Liquidity Spread* are the respective components of the tax-adjusted spread resulting from the transaction-based procedure outlined in Section II.C. *Bond Duration* is the bond's duration. *Bond Age* is the number of years since issuance. *Log(Bond Amount Issued)* is the log of the dollar amount in the bond issue. *Issuer Debt/Revenue* is the ratio of debt outstanding to total revenue, reported annually by the U.S. Census Bureau. *State GDP per Capita* is obtained annually from the U.S. Bureau of Economic Analysis. The rating variables are dummies equal to one if the bond is in that rating category. The missing category includes bonds rated BBB or below and nonrated bonds. *t*-statistics based on standard errors clustered by state and year are reported in brackets. *, **, and *** indicate that the corresponding *p*-values are less than 0.05, 0.01, and 0.001, respectively.

Dependent Variable:	(1) Tax-Adjusted Spread	(2) Liquidity Spread	(3) Default Spread
Bond Duration	14.4*** [14.4]	4.22*** [8.35]	10.2*** [12.5]
Bond Age	3.97*** [10.6]	2.13*** [6.04]	1.84*** [7.99]
Log(Bond Amount Issued)	-2.78 [-1.91]	-1.03 [-1.43]	-1.75 [-1.42]
Issuer Debt/Revenue	37.1* [2.03]	4.49 [1.20]	32.6* [2.09]
State GDP per Capita (\$000s)	-2.14** [-2.61]	-0.19 [-0.98]	-1.95** [-2.75]
AAA-Rated	-107*** [-5.97]	-33.2 [-1.65]	-74.3*** [-4.62]
AA-Rated	-101*** [-6.53]	-30.7 [-1.52]	-70.0*** [-4.42]
A-Rated	-12.1 [-1.06]	-22.5 [-1.12]	10.4 [0.98]
Constant	219*** [9.68]	41.4 [1.87]	177*** [7.14]
<i>R</i> ²	0.36	0.18	0.29
Observations	19,564	19,564	19,564

- Bond duration and bond age matter for both components, which is sensible because longer and older bonds can be both riskier and harder to trade.

Economic message. The decomposition passes an important validation test: fiscal stress and credit quality show up where they should, namely in default spreads.

5.7 Table VII: Comparison of decomposition methods

Read.

- On the state-CDS subsample, the average five-year tax-adjusted spread is 125 bps.
- The transaction-based approach implies a default spread of 92.6 bps and a liquidity spread of 32.4 bps.
- The CDS premium is 98.8 bps, so the CDS-based approach attributes about 84% of the spread to default.
- The average difference between the transaction-based default spread and the CDS premium is only about -6 bps.

Economic message. A very different data source — state CDS — tells essentially the same story. This is one of the most convincing pieces of evidence in the paper.

5.8 Figure 3: Liquidity spreads and the bond-CDS basis

Read.

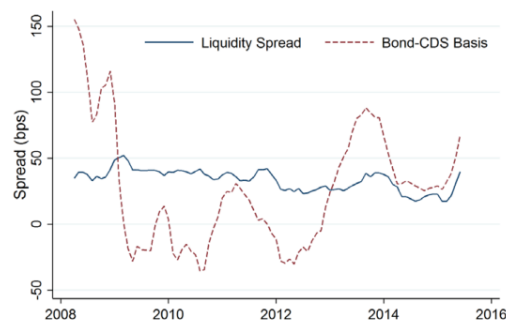
- During and just after the financial crisis, the bond-CDS basis is often 50 to 100 bps above the transaction-based liquidity spread.
- After 2012, the bond-CDS basis declines sharply and sometimes becomes close to zero or negative.

Table VII
Comparison of Decomposition Methods

This table reports summary statistics for municipal bond and CDS spreads at a five-year maturity. The sample includes 936 observations for states with nonmissing CDS premia, a subset of the months between 2008 and 2015. The five-year bond spreads are generated by taking individual bond spreads in a window of one to nine years to maturity, regressing these spreads on time to maturity, and taking the fitted value at five years. Only issuers with trading in at least one bond below and one bond above five years to maturity are included in a given month. *Tax-Adjusted Yield* is the yield adjusted upward with the top statutory income tax rate as in equation (3). *Tax-Adjusted Spread* is the tax-adjusted yield minus the interpolated maturity-matched swap rate. *Default Spread* and *Liquidity Spread* are the respective components of the tax-adjusted spread resulting from the transaction-based procedure outlined in Section II.C. *Default/Tax-Adj.* is the ratio of the default spread to the tax-adjusted spread. *CDS Premium* is the quoted five-year credit default swap premium from Bloomberg. *Tax-Adj.-CDS Basis* is the difference between the tax-adjusted spread and the CDS premium. *CDS/Tax-Adj.* is the ratio of the CDS premium to the tax-adjusted spread. *Default-CDS Basis* is the difference between the default spread from the transaction-based approach and the CDS premium.

	Mean	StDev	P_1	P_{10}	P_{50}	P_{90}	P_{99}
Tax-Adjusted Spread (bps)	125	71.9	24.4	57.4	108	238	383
<i>Transaction-Based Approach</i>							
Default Spread (bps)	92.6	67.1	-8.78	31.0	74.4	190	337
Liquidity Spread (bps)	32.4	15.4	5.47	14.9	30.6	52.0	78.8
Default/Tax-Adj.	0.69	0.20	-0.34	0.48	0.74	0.86	0.93
<i>CDS-Based Approach</i>							
CDS Premium (bps)	98.8	67.4	26.0	33.0	81.0	195	307
Tax-Adj.-CDS Basis (bps)	26.3	53.1	-106	-38.8	24.4	89.4	195
CDS/Tax-Adj.	0.84	0.45	0.23	0.35	0.76	1.43	2.59
Default-CDS Basis (bps)	-6.13	53.8	-150	-74.5	-4.20	55.8	149

spreads in Section III.A.2, this approach is associated with two potential shortcomings. First, the sensitivity of the tax-adjusted spread could be biased by



3. **Liquidity spreads and bond-CDS basis.** This figure presents time-series plots of spreads from the transaction-based and CDS-based approaches. The sample is restricted to states with nonmissing CDS premia, a subset of the months between 2008 and 2015. Each line is an equal-weighted average over four-month rolling windows. The solid line is the estimated liquidity spread from the transaction-based approach. The dashed line is the bond-CDS basis, or the difference between the tax-adjusted spread and the CDS premium, which represents the liquidity premium from the CDS-based approach. (Color figure can be viewed at wileyonlinelibrary.com)

the CDS-based approach, it ranges from 62% to 104%. The liquidity

- The transaction-based liquidity spread is much smoother than the bond-CDS basis.

Economic message. This figure suggests that the transaction-based approach may understate absolute liquidity premia in crisis periods. Even so, outside crisis conditions the two approaches line up well, and default still dominates overall.

5.9 Table IX: Pre-refunded bonds

Read.

- Within 12 months of the event, the average tax-adjusted spread falls from about 211 bps before pre-refunding to about 103 bps after.
- In the specification with bond and month fixed effects, the pre-refunding coefficient is about -86.5 bps.
- The decline is even larger for lower-rated bonds, especially A-rated bonds.

Economic message. When default risk is effectively removed, spreads fall a lot. That makes it very hard to argue that liquidity is the dominant component of municipal yields.

5.10 Table X: Implied default risk premium

Read.

- Using the paper's preferred default spread estimates, the implied annual risk-neutral default probability is many multiples of the physical default probability.
- Under $LGD = 50\%$, the implied q/p multiple is between about 108 and 126 across the paper's decomposition methods.
- Even under the conservative $LGD = 100\%$ case, the multiple is still roughly 54 to 63.
- These multiples are far above corresponding estimates for investment-grade corporate bonds.

Economic message. The paper's big surprise is not just that default risk matters, but that the *price* of municipal default risk is extraordinarily high relative to realized default frequencies.

Table IX
Pre-Refunded Bonds

This table reports statistics and regressions on pre-refunded bonds. Panel A reports summary statistics. *Years to Call* is the number of years from the trade date to the first call date. *Tax-Adjusted Spread* is the difference between the tax-adjusted yield-to-call and the interpolated maturity-matched swap rate. Rows labeled $m > -12$ and $m < 12$ include only trades within 12 months of the pre-refunding month. Panel B reports regressions of tax-adjusted spreads within 12 months of pre-refunding on an indicator for whether the bond is pre-refunded and control variables. *Pre-Refunded* is a dummy equal to one if the trade occurs after pre-refunding. *Pre-Refunded**AA and *Pre-Refunded**A are interactions of *Pre-Refunded* and indicators for whether the bond is rated AA or A. The omitted category is AAA-rated bonds. *Rating Category* includes rating category dummies. *Bond Controls* include years to call and log issue amount. *t*-statistics based on standard errors clustered by bond and month are in brackets. *, **, and *** indicate that the corresponding *p*-values are less than 0.05, 0.01, and 0.001, respectively.

Panel A: Summary Statistics						
	Mean	StDev	p10	p50	p90	Obs.
<i>Before Pre-Refunding</i>						
Years to Call	4.06	2.28	1.60	3.40	7.63	2,004
Tax-Adjusted Spread (bps)	222	138	75.1	196	398	2,004
Tax-Adjusted Spread (bps) ($m < 12$)	211	133	69.2	186	384	1,462
Tax-Adj. Spread (bps) (A-Rated, $m < 12$)	245	139	102	217	406	48
<i>After Pre-Refunding</i>						
Years to Call	2.46	1.83	0.71	1.82	5.26	2,632
Tax-Adjusted Spread (bps)	98.3	95.6	15.7	85.4	200	2,632
Tax-Adjusted Spread (bps) ($m < 12$)	103	84.1	26.8	90.6	196	1,016
Tax-Adj. Spread (bps) (A-Rated, $m < 12$)	105	76.4	45.2	86.5	202	23
Panel B: Regressions						
Tax-Adj. Spread	(1)	(2)	(3)	(4)	(5)	(6)
Pre-Refunded	-107*** [-14.7]	-83.6*** [-8.51]	-86.5*** [-8.99]	-111*** [-12.6]	-87.2*** [-7.80]	-95.9*** [-9.17]
Pre-Refunded*AA				11.3 [1.22]	11.3 [1.01]	23.1* [2.56]
Pre-Refunded*A				-59.6* [-2.06]	-114** [-3.23]	-122*** [-4.82]
Rating Category	X	X	X	X	X	X
Bond Controls	X	X	X	X	X	X
Bond FES		X	X		X	X
Mean VIF			v			v

Table X
Implied Default Risk Premium

This table reports estimates of implied default risk premium multiples based on the default spread estimates in the previous tables. *Default Spread* is the average or baseline specification for each approach based on the full sample. *ABX (2014)* reports the average default spread for the period 1996 to 2013 from Ang, Bhansali, and Xing (2014). The following formula is used to compute the risk-neutral default probability: $\gamma = q * LGD$, where γ is the default spread, q is the risk-neutral default probability, and *LGD* is the expected loss given default. Three scenarios are considered for loss given default, two based on outcomes from recent municipal bankruptcies and one extreme scenario. The risk-neutral to physical default probability ratio q/p is a multiple reflecting the default risk premium (Berndt et al. (2005)). The physical default probability is assumed to be 0.016%, the annualized default rate for rated municipal bonds, based on the 0.08% historical five-year cumulative default rate for 1970 to 2014 reported by Moody's.

Method	Default Spread	LGD = 50%		LGD = 80%		LGD = 100%	
		<i>q</i>	<i>q/p</i>	<i>q</i>	<i>q/p</i>	<i>q</i>	<i>q/p</i>
Transaction	101 bps	2.02%	126	1.26%	78.8	1.01%	63.1
CDS	98.8 bps	1.98%	124	1.24%	77.5	0.99%	61.8
Pre-Refunding	86.5 bps	1.73%	108	1.08%	67.5	0.87%	54.1
ABX (2014)	45.4 bps	0.91%	56.9	0.56%	35.0	0.45%	28.4

6 Short summary

- The paper asks whether municipal bond spreads mainly reflect liquidity or default risk after adjusting for taxes.
- It studies a large sample of GO municipal bonds using MSRB transaction data, Bloomberg bond characteristics, fiscal data, state GDP, state CDS, and pre-refunding events.
- The main sample is cleaned aggressively to remove insurance, callable bonds, taxable bonds, AMT bonds, new issues, and bonds close to maturity.
- A transaction-based decomposition assigns about 74% of the average tax-adjusted spread to default risk and about 26% to liquidity.
- A CDS-based decomposition on state issuers assigns an even larger share — about 84% — to default risk.
- A within-bond pre-refunding design shows spreads fall by about 86.5 bps when default risk is removed, implying a conservative lower bound of 41% for the default share.
- Fiscal variables and ratings load mainly on default spreads, not on liquidity spreads, which supports the credibility of the decomposition.
- The implied municipal default risk premium is much larger than in the corporate bond market.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that, once municipal bond yields are adjusted for their tax advantage, the majority of the remaining spread is compensation for default risk rather than illiquidity. Across

several complementary methods, default accounts for roughly three-quarters or more of the average spread, while liquidity plays a secondary role.

7.2 Economic intuition

Municipal bonds are hard to trade, but most investors in this market are buy-and-hold households, so large transaction costs do not automatically translate into large liquidity discounts. By contrast, even a very small probability of fiscal distress can command a large spread if the market price of default risk is high and if risk-bearing capital is limited by tax-driven segmentation and retail ownership.

7.3 Takeaway

The paper's central message is sharp: municipal bond spreads are mostly about priced credit risk, not about trading frictions. The main unresolved question is therefore not whether municipal bonds are illiquid — they are — but why the market demands such an unusually large premium for bearing municipal default risk.